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Monte Carlo Simulation for Buffon's Needle: Classic and Concentric Circle Variants

CmpE 49G
FUNDAMENTALS OF PARTICLE-BASED SIMULATIONS

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1 Introduction

Buffon's Needle connects geometric probability with π. We study two geometries with the same parameters (needle length L and spacing scale D) under the assumption $L < D$:

- **Parallel lines (classic):** equally spaced lines with distance D.
- **Concentric circles (variant):** circles with radii $r_k = kD$.

Both are simulated with vectorized NumPy and compared to the analytical prediction $P = 2L/(\pi D)$.

2 Theory (Classic Buffon)

Let $x \sim U(0, D/2)$ be the distance from the needle center to the nearest line and $\theta \sim U(0, \pi/2)$ the orientation angle. A crossing occurs iff

$$x \le \frac{L}{2}\sin\theta.$$

Integrating over the uniform (x, θ) distribution gives

$$P(\text{cross}) = \frac{2L}{\pi D} \qquad (L < D).$$

3 Concentric Circles Variant

For circles with radii $r_k = kD$ and $L < D$, a needle can intersect at most one circle. A standard integral-geometry result implies the crossing probability matches the classic case:

$$P(\text{cross any circle}) = \frac{2L}{\pi D} \qquad (L < D).$$

We treat this as the analytical target and verify it numerically.

Formal Proof via the Cauchy–Crofton Formula

Let the needle centre be uniform in a square of side $S \gg D$ and angle $\Theta \sim U(0, 2\pi)$. The *Cauchy–Crofton formula* from integral geometry states: for any rectifiable plane curve of total length ℓ inside a region of area A, the expected number of intersections with a randomly placed needle of length L is

$$\mathbb{E}[\#\,\text{intersections}] = \frac{2L\,\ell}{\pi A}.$$

Step 1 – Total curve length. The circles with radii kD ($k = 1, \ldots, K$, where $K \approx S/(D\sqrt{2})$) that lie inside the square have combined length

$$\ell = \sum_{k=1}^{K} 2\pi k D = \pi D\, K(K+1) \approx \frac{\pi S^2}{2D}.$$

Step 2 – Apply the formula. With $A = S^2$:

$$\mathbb{E}[\#\,\text{intersections}] = \frac{2L \cdot \dfrac{\pi S^2}{2D}}{\pi S^2} = \frac{2L}{\pi D}.$$

Step 3 – At most one crossing. Because $L < D$ the needle straddles at most one circle, so

$$\mathbb{E}[\#\,\text{intersections}] = P(\text{cross any circle}).$$

Conclusion.

$$\boxed{P(\text{cross any circle}) = \frac{2L}{\pi D}.} \qquad \square$$

4 Simulation Method

For each geometry we run $N \in \{10^2, 10^3, 10^4, 10^5, 10^6\}$ trials for three parameter pairs $(L, D) \in \{(1,2),(2,3),(3,5)\}$ with a fixed RNG seed (42).

- **Parallel lines:** sample x and θ, then apply the crossing condition.
- **Concentric circles:** sample a needle center uniformly in a large square of side $200D$ and a uniform angle in $[0, 2\pi)$, then test for an intersection with the relevant circle radii near the needle.

5 Results

Figure 1 and Figure 2 show convergence of the Monte Carlo estimates with increasing N. Table 1 summarizes the $N = 10^6$ estimates.

Case	L	D	P_{theory}	$\hat{P}$ (lines)	$\hat{P}$ (circles)
1	1.0	2.0	0.31830989	0.317456	0.317924
2	2.0	3.0	0.42441318	0.423831	0.423841
3	3.0	5.0	0.38197186	0.381245	0.381628

Table 1: Summary at $N = 10^6$ (seed=42). Both geometries converge to $2L/(\pi D)$ within typical Monte Carlo sampling error.

6 Convergence and Error Analysis

Detailed Convergence Tables

Tables 2–4 report estimated probabilities and absolute errors for every (N, L, D) combination across both geometries. The theoretical value for each case is $P_{th} = 2L/(\pi D)$.

Error Scaling Plots

Figures 3 and 4 show the absolute error $|\epsilon|$ on a log–log scale. A dashed reference line with slope $-\frac{1}{2}$ traces the theoretical $O(N^{-1/2})$ Monte Carlo rate. All three (L, D) cases follow this slope closely for both geometries, confirming:

- No systematic algorithmic bias: deviations are random noise, not a trend.
- Consistent scaling: halving the error requires quadrupling N, as expected.
- At $N = 10^6$ the typical error is $O(10^{-4})$, corresponding to sub-0.1% relative error.

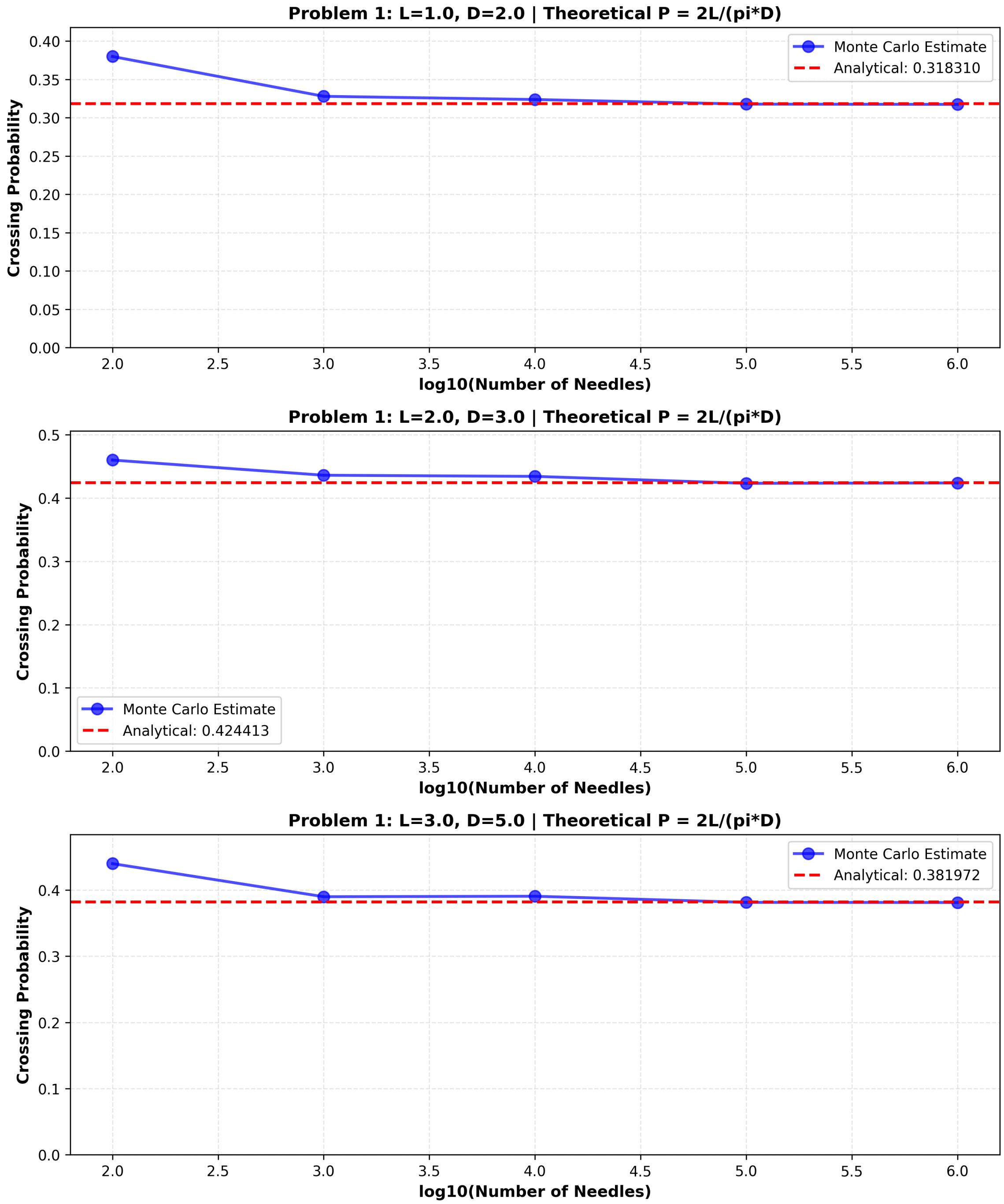

Figure 1: Problem 1: convergence of Monte Carlo estimates to $2L/(\pi D)$ for three (L, D) cases.

7 Conclusion

Across all tested (L, D) pairs and sample sizes up to 10^6, Monte Carlo estimates converge to the analytical probability $2L/(\pi D)$. The concentric-circles geometry empirically matches the same probability as the classic parallel-line setup, and the error decreases with the expected $O(N^{-1/2})$ Monte Carlo rate.

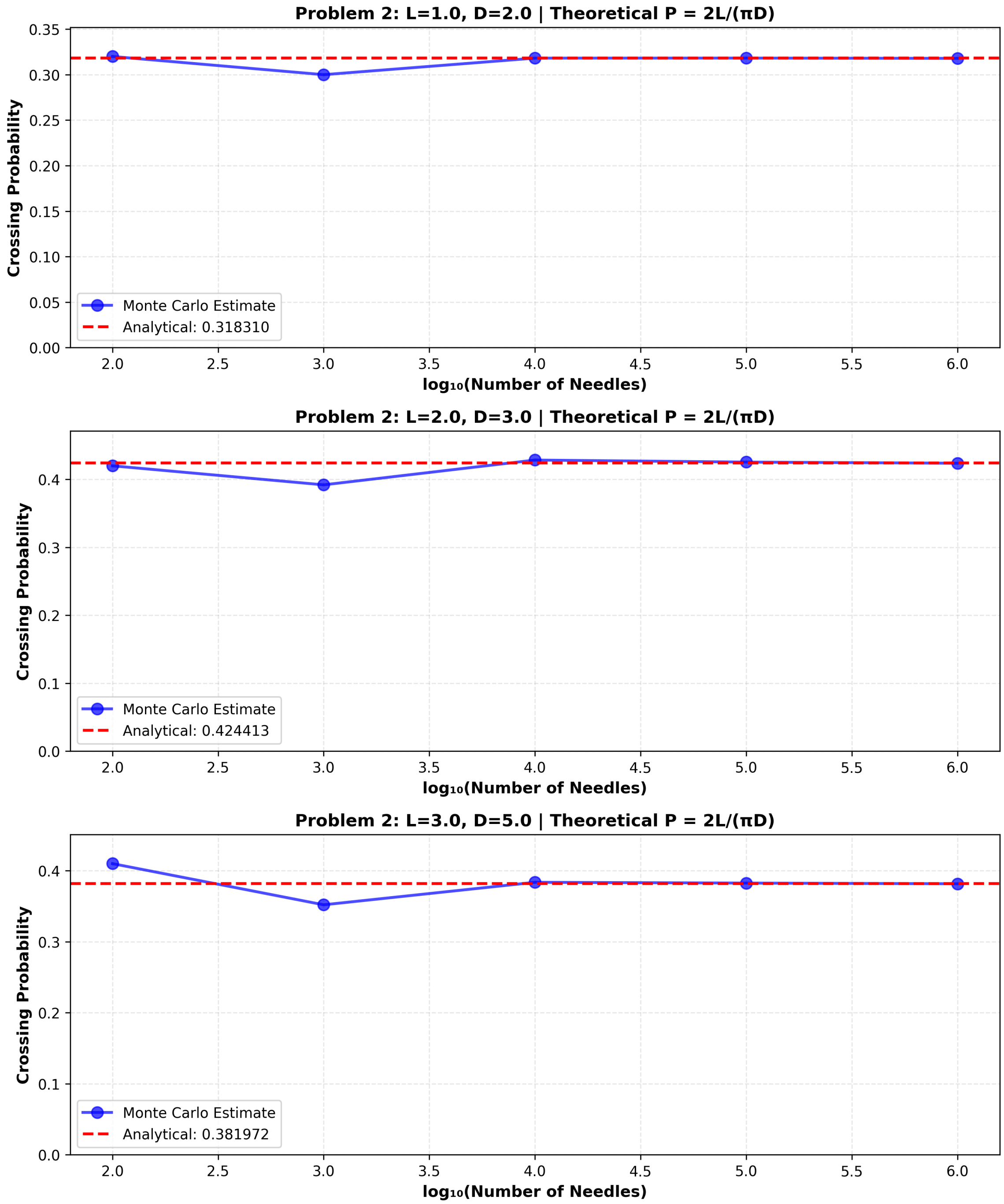

Figure 2: Problem 2: convergence for concentric circles. Empirically, behaviour matches the classic case.

8 AI Usage Declaration

What AI Was Used For

- Initial code architectural structure and function organisation
- Documentation template generation (README, report skeleton)

	Parallel Lines		Concentric Circles	
N	$\hat{P}$	$\|\epsilon\|$	$\hat{P}$	$\|\epsilon\|$
10^2	0.38000	0.06169	0.32000	0.00169
10^3	0.32800	0.00969	0.30000	0.01831
10^4	0.32380	0.00549	0.31820	0.00011
10^5	0.31780	0.00051	0.31825	0.00006
10^6	0.31746	0.00085	0.31792	0.00039

Table 2: Case 1: $L = 1.0$, $D = 2.0$, $P_{th} = 0.31831$.

	Parallel Lines		Concentric Circles	
N	$\hat{P}$	$\|\epsilon\|$	$\hat{P}$	$\|\epsilon\|$
10^2	0.46000	0.03559	0.42000	0.00441
10^3	0.43600	0.01159	0.39200	0.03241
10^4	0.43420	0.00979	0.42850	0.00409
10^5	0.42322	0.00119	0.42551	0.00110
10^6	0.42383	0.00058	0.42384	0.00057

Table 3: Case 2: $L = 2.0$, $D = 3.0$, $P_{th} = 0.42441$.

	Parallel Lines		Concentric Circles	
N	$\hat{P}$	$\|\epsilon\|$	$\hat{P}$	$\|\epsilon\|$
10^2	0.44000	0.05803	0.41000	0.02803
10^3	0.39000	0.00803	0.35200	0.02997
10^4	0.39070	0.00873	0.38370	0.00173
10^5	0.38145	0.00052	0.38261	0.00064
10^6	0.38125	0.00073	0.38163	0.00034

Table 4: Case 3: $L = 3.0$, $D = 5.0$, $P_{th} = 0.38197$.

- Code style and professional formatting suggestions

References

[1] Buffon, G.-L. L. (1777). "Essai d'arithmétique morale." *Histoire Naturelle, Générale et Particulière, Supplement 4*.

[2] Kendall, M. G., & Moran, P. A. P. (1963). *Geometrical Probability.* Hafner Publishing.

[3] Ross, S. M. (2012). *Simulation* (5th ed.). Academic Press.

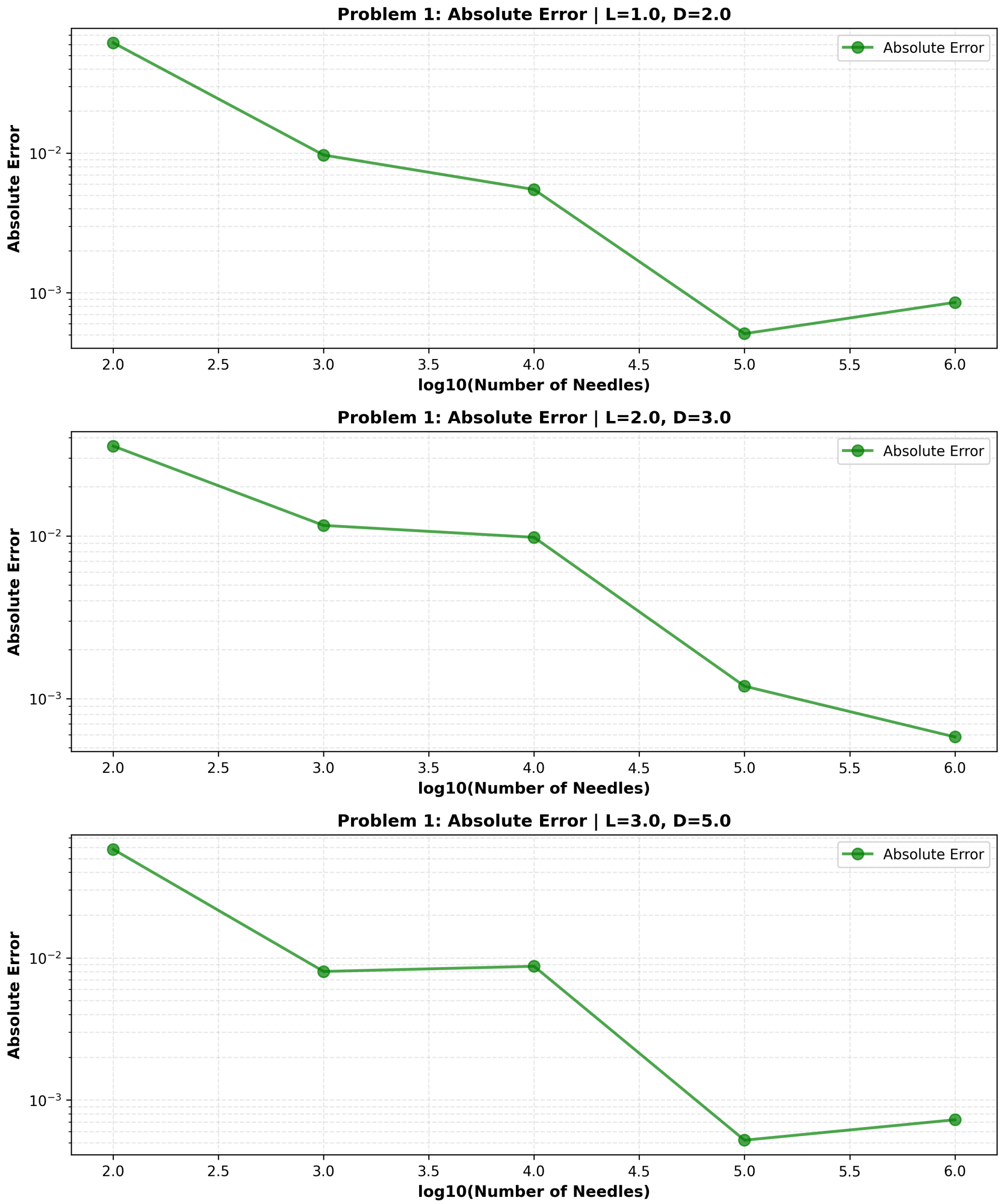

Figure 3: Problem 1 (parallel lines): absolute error vs. N on log–log axes. Dashed line: reference slope $-1/2$.

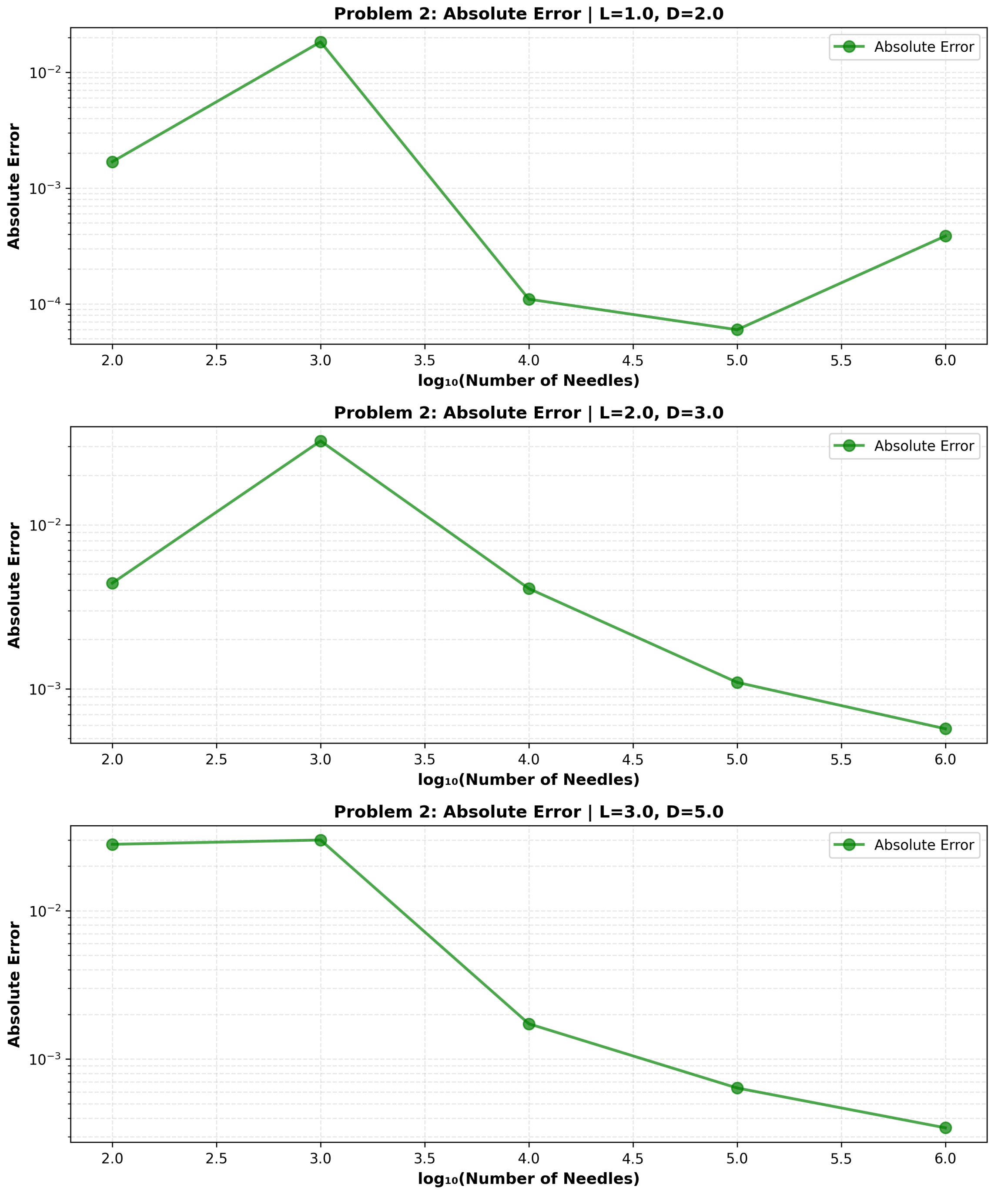

Figure 4: Problem 2 (concentric circles): absolute error vs. N. The $O(N^{-1/2})$ convergence rate matches Problem 1.